SHUNAN SHENG

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Social Work Building, 1255 Amsterdam Avenue, New York, NY 10027

EDUCATION

Ph.D. in Statistics, Columbia University *Sept 2022 - May 2027 (expected)*
Supported by the Bloomberg Quantitative Finance Ph.D. Fellowship 2024 - Present
Advisors: Prof. Marcel Nutz and Prof. Alberto González-Sanz

B.S. in Mathematical Sciences, Nanyang Technological University, Singapore *Aug 2018 - May 2022*
Awarded the Senior Middle 2 Scholarship
Advisor: Prof. Ariel Neufeld

RESEARCH INTERESTS

- Optimal Transport, Robust Finance, Variational Inference, Sampling

PUBLICATIONS & PREPRINTS

González-Sanz, A., Mordant, G., & **Sheng, S.** (2026). Empirical Optimal Transport Potentials: Fast Rates and a Functional Central Limit Theorem. arXiv preprint arXiv:2608.00649.

González-Sanz, A., **Sheng, S.**, Wu, B., & Avella Medina, M. (2026). The Influence Function of Transport-based Quantiles. arXiv preprint arXiv:2607.19080.

Liang, B., Nutz, M., **Sheng, S.**, & Tissot-Daguette, V. (2026). Bid–Ask Martingale Optimal Transport. arXiv preprint arXiv:2603.24605. *(Major revision at Finance and Stochastics)*

Sheng, S., Wu, B., Zhu, B., Chewi, S., & Pooladian, A. A. (2025). Theory and Computation for Structured Variational Inference. arXiv preprint arXiv:2511.09897.

Sheng, S., Wu, B., & González-Sanz, A. (2025). Mode Collapse of Mean-Field Variational Inference. arXiv preprint arXiv:2510.17063.

Sheng, S., Wu, B., González-Sanz, A., & Nutz, M. (2025). Stability of Mean-Field Variational Inference. arXiv preprint arXiv:2506.07856. *(Major revision at Probability Theory and Related Fields)*

González-Sanz, A., & **Sheng, S.** (2024). Linearization of Monge-Ampère Equations and Statistical Applications. arXiv preprint arXiv:2408.06534.

Sheng, S., Xiang, Q., Nevat, I., & Neufeld, A. (2024). Binary spatial random field reconstruction from non-Gaussian inhomogeneous time-series observations. Journal of the Franklin Institute, 361(2), 612–636.

Ren, J., Yu, C., **Sheng, S.**, Ma, X., Zhao, H., Yi, S., & Li, H. (2020). Balanced meta-softmax for long-tailed visual recognition. Advances in Neural Information Processing Systems, 33, 4175–4186.

INDUSTRY RELATED EXPERIENCE

Bloomberg L.P. *Summers 2025, 2026*
Quantitative Researcher (Internship)

National University of Singapore, Singapore *May 2021 - Aug 2021*
Undergraduate Researcher (Research in Industrial Projects for Students, RIPS 2021)

SenseTime Research, Singapore *Nov 2019 - Jun 2020*
Algorithm Researcher (Internship)

AWARDS & HONORS

Quantitative Finance Ph.D. Fellowship, Bloomberg L.P., New York — *Sept 2024 - Present*

Singapore National Academy of Science (SNAS) Award, Singapore — *Jun 2022*

Senior Middle 2 Scholarship, NTU & Ministry of Education (MOE), Singapore — *Aug 2017 - Jun 2022*

Stonehage Fleming/Marais Prize, Simon Marais Mathematical Competition — *Dec 2019*

SERVICE

Reviewer: Journal of the American Statistical Association, SIAM Journal on Optimization, SIAM Journal on Mathematics of Data Science, ICLR 2026, Statistics and Probability Letters, Computational Statistics and Data Analysis

Organizer: Minghui Yu Memorial Conference 2025, Columbia + NYU + Princeton Ph.D. Joint Seminar 2026

TEACHING

Teaching Assistant at Columbia University

Time Series Modeling (Master/undergrad-level, GR5263/GU4263) — *Fall 2024*

Nonparametric Statistics (Master-level, GR5222) — *Spring 2024*

Introduction to Statistics with Calculus (Undergrad-level, UN1201) — *Fall 2023*

Introduction to Statistical Reasoning (Undergrad-level, UN1001) — *Summer 2023*

Statistical Machine Learning (Master-level, GR5241) — *Spring 2023*

Applied Linear Regression (Undergrad-level, UN2103) — *Fall 2022*

TALKS & POSTERS

2nd INFORMS Conference on Financial Engineering and Fintech (invited talk) — *Nov 2026 (scheduled)*

Columbia + NYU + Princeton Ph.D. Joint Seminar (invited talk), Princeton University — *Oct 2026*

Student Seminar at the Department of Statistics (invited talk), Columbia University — *Oct 2026*

ICORS 2026 International Conference on Robust Statistics (invited talk) — *July 2026*

CIB Data Science Seminar (virtual), Wells Fargo Corporate & Investment Banking — *May 2026*

Workshop on Robust Statistics (poster), Columbia University — *April 2026*

Minghui Yu Memorial Conference 2026 (contributed talk), Columbia University — *April 2026*

Bloomberg Quant (BBQ) Seminar (invited talk), Bloomberg L.P. — *Mar 2026*

ICSDS 2025 International Conference on Statistics and Data Science (contributed talk) — *Dec 2025*

Minghui Yu Memorial Conference 2025 (contributed talk), Columbia University — *April 2025*

Optimization and Statistical Learning Workshop (poster), Columbia University — *April 2025*

Statistics and Optimal Transport Workshop (poster), Columbia University — *Mar 2025*

ICSDS 2024 International Conference on Statistics and Data Science (poster) — *Dec 2024*

FDS Conference: Recent Advances and Future Directions for Sampling (poster), Yale University — *Oct 2024*